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Chapter 1

The Fundamentals of Gravitational Wave Interferometers

1.1 Introduction

On the 14th of September 2015, the detection of a gravitational wave emanating from two coalescing black holes signalled the beginning of a new era for astronomy [1]. The information carried by a gravitational wave is undisturbed by the matter between the observer and the astrophysical body that is emitting them, thus it gives astronomers valuable information which cannot be obtained from the electromagnetic radiation emitted by that body. Gravitational wave astronomy also allows for direct observations of objects that do not emit electromagnetic radiation, e.g. black holes. Therefore, it is important that large numbers of gravitational wave signals are measured to develop our understanding of the universe.

However, gravitational waves are incredibly weak and the detection of them is a science in itself. This chapter will outline how a gravitational wave interferometer, with a particular emphasis on LIGO [2], can be used to detect these signals and the fundamental noise sources these detectors face. While there are other important sources of noise, the suppression of quantum noise is crucial to achieving the sensitivity required to detect gravitational waves; therefore, quantum noise is the focus of this chapter.

1.2 The Astrophysical Origins of Gravitational Waves

Gravitational waves produced by inspiraling pairs of massive objects, such as binary black holes (BBH) (e.g. [1, 3–5]), binary neutron stars (BNS) [6, 7], and a source which is either a BBH or a binary consisting of a black hole and a neutron star (NSBH) [8], have been detected. Every confirmed gravitational wave signal to date is summarised in [9]. The waveform of the gravitational radiation emitted from these systems is known as a chirp because the frequency of the gravitational wave signal increases as the two objects get closer to merging [10]. The cut-off frequency is determined by the radius of the objects that will merge; for the signals which have been measured, the cut-off frequencies were between 100 Hz and 10 kHz. A typical gravitational wave strain amplitude of these types of systems, all of which have a total mass of $10M_{\odot} - 100M_{\odot}$ and are of the order 100 Mpc to 10 Gpc away from Earth, is $h \sim 10^{-21} - 10^{-23}$.

Einstein’s general theory of relativity accurately describes the gravitational waves measured with the LIGO and Virgo interferometers [11]. Gravitational waves are

emitted from mass quadrupole systems (see e.g. [12] or [13]). As the coupling between mass and the curvature of space-time is weak, gravitational waves are only observable in systems that have large masses moving at relativistic velocities.

Other astrophysical phenomena will emit gravitational waves. Asymmetries in pulsars [14,15] and magnetars [16] will cause them to emit a gravitational wave with a constant frequency. When the core of a star collapses as it goes supernova, a burst of gravitational radiation may be emitted [17,18]. Burst signals may also be generated immediately after a BNS merges [19]. Weaker sources that future gravitational wave interferometers may detect include: a stochastic background level of gravitational waves due to the merging of compact binaries from all over the universe [20], dark matter [21,22], and sources that existed during the early universe [23,24].

To improve statistical models of the systems that emit gravitational waves, the number of detections needs to be increased. To detect continuous wave signals, the signal to noise ratio (SNR) improves with the total observing time of the detector. With longer observation times, it is more likely that an exotic event will be measured. For these reasons, gravitational wave detectors are operated for year long timescales with minimal interruption. These are known as observation runs. Between observation runs, the detectors undergo hardware upgrades. The previous, current and future observation runs are summarised in Table 1.1; a more comprehensive overview is found in [25].

Observation Run	Dates	Detections	Detectors
O1	12/09/2015 - 19/01/2016	3	aLIGO
O2	30/11/2016 - 25/08/2017	8	aLIGO, Virgo
O3a	01/04/2019 - 01/10/2019	39	aLIGO, Virgo
O3b	01/11/2019 - 30/04/2020	23	aLIGO, Virgo
O4	2021	-	aLIGO, Virgo, KAGRA
O5	2025	-	LIGO A+, Advanced Virgo, KAGRA, LIGO India

Table 1.1: Summary of observation runs [25] for LIGO, Virgo and KAGRA [2,26,27].

1.3 Measuring Gravitational Waves with an Interferometer

Gravitational waves are propagating longitudinal perturbations in the curvature of space-time, and can be split into a basis set containing two polarisations, plus and cross (e.g. [12] or [13]). The effect these waves have, in the local Lorentz frame, on free test masses on Earth is to move them by a small amount ¹.

To detect plus polarised gravitational waves, the end mirrors of a Michelson interferometer can be used as the test masses. These mirrors are referred to as *end test masses* (ETMs). A suspended mass will behave in the same way as a free mass when it is driven at above its resonant frequency, so the ETMs need to be

¹In the local Lorentz frame of the beam splitter, for any feasible Earth based detector, the effect of a gravitational wave on the wavelength of the light is negligible compared to its effect on the positions of the test masses due to the strong equivalence principle: All non-gravitational laws of physics which can be expressed in a special relativistic form have the same form in a local Lorentz frame (e.g. [12]). In the local Lorentz frame, the metric has the form $g_{\alpha\beta} = \eta_{\alpha\beta} + \mathcal{O}(h)$, where $\eta_{\alpha\beta}$ is the Minkowski space-time metric and $\mathcal{O}(h)$ is a perturbation of the order $\frac{4\pi L}{\lambda_{\text{EM}}}^2 h$. The Minkowski space-time metric is flat, so in the local Lorentz frame the metric is, essentially, flat and the gravitational waves cause the test masses to move by $\Delta L = hL$. Maxwell's equations are modified by a term that is of the order $\frac{\delta\lambda_{\text{EM}}}{\lambda_{\text{EM}}} \sim \frac{4\pi L}{\lambda_{\text{EM}}}^2 h$, and so if $\frac{4\pi L}{\lambda_{\text{EM}}}$ is small, as it would be for an Earth based interferometer, then the effect of the gravitational wave on the light is much smaller than the effect of the gravitational wave on the test masses.

suspended [28]. A sketch of a gravitational wave interacting with a Michelson interferometer is shown in Figure 1.1. For test masses at a distance L away from the origin of the local Lorentz frame, the distance they move, ΔL , due to an incoming plus polarised gravitational wave of strain h is

$$\Delta L = Lh. \quad (1.1)$$

Interferometers are sensitive to arm length changes that are on the order of one wavelength, λ_{EM} , of the light they are operated with. The differential phase accumulated by the light in the interferometer's arms, $\Delta\phi$, is

$$\Delta\phi = \frac{4\pi}{\lambda_{\text{EM}}} Lh. \quad (1.2)$$

The wavelength current interferometers operate with, $\lambda_{\text{EM}} = 1064 \text{ nm}$, was selected because test masses with low noise and mirror coatings with low loss and low thermal noise can be fabricated for this wavelength [29].

The gravitational wave strain is converted into intensity fluctuations of light at the *antisymmetric* port (see Figure 1.1) due to the two light fields from the arms interfering at the beam splitter. When on the *dark fringe*, there is a small amount of light at the antisymmetric port for the signal light to beat with and so the variation in the intensity of the light depends linearly on the gravitational wave strain. The interferometer is set to operate on the dark fringe by introducing a small static differential offset, ϕ_0 , in the differential arm length. This technique is known as DC readout [30, 31]. At the beam splitter, the electric field, Ψ_{AS} , emerging from the antisymmetric port due to the interference of the beams returning from the ETMs,

Ψ_y and Ψ_x , with differential phase $\phi = \phi_0 + \Delta\phi$ is

$$\Psi_{AS} = \frac{\Psi_y}{\sqrt{2}} - \frac{\Psi_x}{\sqrt{2}}, \quad (1.3)$$

$$= \sqrt{\frac{I_0}{2}} \left(\frac{e^{i(\phi_0 + \Delta\phi)}}{\sqrt{2}} - \frac{e^{-i(\phi_0 + \Delta\phi)}}{\sqrt{2}} \right), \quad (1.4)$$

where I_0 is the power of the beam injected into the beam splitter by the laser. The intensity of the beam at the antisymmetric port, I_{AS} , is

$$I_{AS} = \Psi_{AS}^* \Psi_{AS}, \quad (1.5)$$

$$\simeq I_0 \phi_0^2 + 2I_0 \phi_0 \Delta\phi, \quad (1.6)$$

$$= I_0 \phi_0^2 + 2I_0 \phi_0 \frac{4\pi}{\lambda_{EM}} Lh. \quad (1.7)$$

In the absence of a gravitational wave signal, the DC light level at the antisymmetric port is related to the static offset by $\phi_0 = \sqrt{\frac{\overline{I_{AS}}}{I_0}}$, where $\overline{I_{AS}}$ is the average of I_{AS} . As we are interested in a fluctuating signal, the DC offset from Equation (1.7) can be removed, allowing for the signal at the antisymmetric port to be written as

$$I_{AS} = 2\sqrt{I_0 \overline{I_{AS}}} \frac{4\pi}{\lambda_{EM}} Lh. \quad (1.8)$$

Therefore, it is possible to measure the differential motion of the ETMs due to a gravitational wave's strain by placing a photodiode in the beam emerging from the antisymmetric port of a Michelson interferometer.

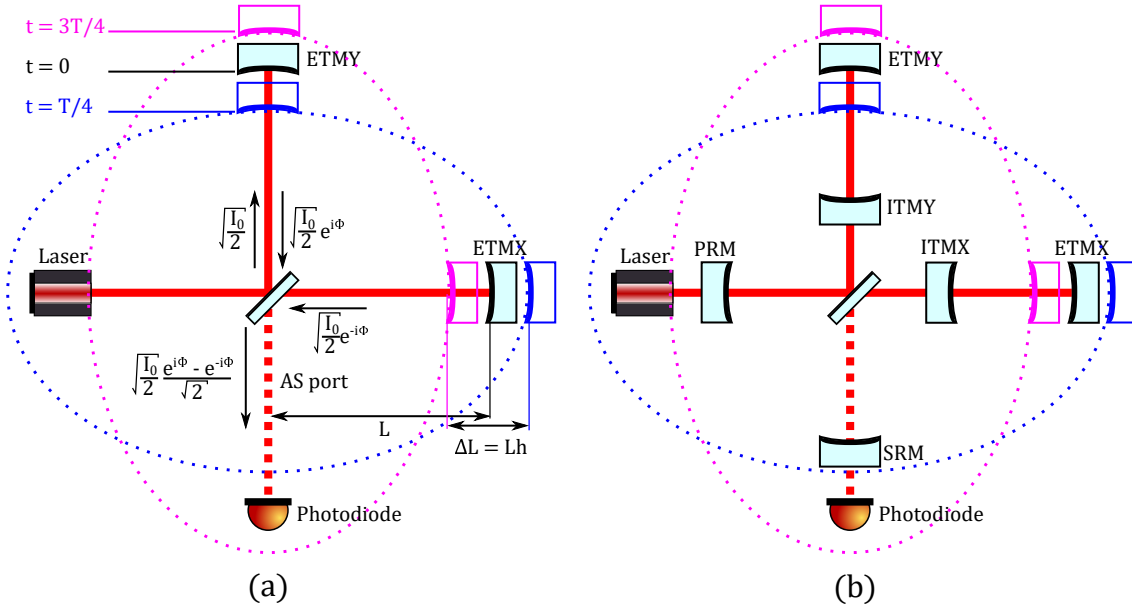


Figure 1.1: (a) A sketch of the effect a gravitational wave has on a Michelson interferometer. The positions of the test masses, ETMX and ETMY, at different points during the gravitational wave's period is shown by the blue and pink lines. The difference in phase shift between the fields in the two arms affects the intensity of the light at the AS port. By detecting this light with a photodiode, the gravitational wave strain h can be determined. (b) To reach the sensitivity needed to detect gravitational wave, several mirrors need to be added to the basic Michelson interferometer: ITMX and ITMY boost the signal generated in the arms, the PRM increases the power at the beam splitter, and the SRM can be used to shape the quantum noise of the interferometer.

1.3.1 Using Fabry-Perot Cavities in the Arms of a Michelson Interferometer to Enhance Its Sensitivity

The optimal arm length of the interferometer is determined by $1/4$ of the wavelength of a gravitational wave as this gives the maximal $\Delta\phi$. Gravitational wave signals from BBH and BNS have frequencies of ~ 100 Hz; therefore, the wavelength of the gravitational wave will be of the order ~ 3000 km. It is not possible to construct a Michelson interferometer with ~ 1000 km arms on Earth.

The sites that were available to construct the LIGO detectors allow for 4 km arms. To reach the optimal arm length on a 4 km site, the number of times that the light bounces off of the ETMs, B , needs to be increased. Numerically, B should be ~ 250 . This means that Equation (1.2) becomes

$$\Delta\phi = \frac{4\pi}{\lambda_{\text{EM}}} BLh. \quad (1.9)$$

Fabry-Perot cavities are used to increase the number of times light bounces off of the ETMs². Cavities are formed by placing an *initial test mass* (ITM) in each arm near the beam splitter. When a photon enters a cavity, it remains in the cavity for an average number of bounces that is determined by the reflectivity of the mirrors. Interferometers measure $\Delta\phi_{\text{EM}} = \frac{d\phi_{\text{EM}}}{d\phi_{\text{ETM}}} \Delta\phi_{\text{ETM}}$, so when a Fabry-Perot cavity is near or on resonance, the Fabry-Perot cavity enhances the detectors sensitivity as the slope $\frac{d\phi_{\text{EM}}}{d\phi_{\text{ETM}}}$ is much greater for a Fabry-Perot than it is for a mirror. This is

²Herriott delay lines were considered as a means of achieving more bounces from the ETMs, however due to difficulty in manufacturing suitable mirrors which do not scatter light, thus generating noise, delay line interferometers fell out of favour. In principle, both delay lines and Fabry-Perot cavities would perform equally well due to the Mizuno limit [32].

illustrated in Figure 1.2. The exact way this happens is dependent of the frequency of the gravitational wave and the reflectivity of the mirrors, with the arm cavity acting as a low pass filter (e.g. see [32, Chapter 3], [33, Appendix D], [34] or [35]).

1.4 Sources of Noise in Gravitational Wave Interferometers.

As the strain induced by a gravitational wave causes a minuscule change in the phase accumulated by light in the arm cavities, the challenge of detecting gravitational waves lies in suppressing the noise below this signal. From Equation (1.9), for a gravitational wave strain of $h_{\text{rms}} = 10^{-23}$ interacting with LIGO, the beams interfering at the beam splitter will have a difference in phase of $\sim 10^{-11}$ rad.

1.4.1 Laser Power and Shot Noise

The shot noise associated with detecting a coherent state, such as that created by the laser, can limit the sensitivity of the interferometer. To reduce the level of shot noise, the energy, E , of the laser needs to be high. The uncertainty in energy, ΔE , that associated with a measurement of the time between two events, ΔT , is determined by the time-energy uncertainty principle, $\Delta E \Delta T \geq \hbar/2$. The uncertainty in the phase, $\Delta \phi$, of a light beam with angular frequency ω_{EM} is related to the measurement time by $\Delta \phi = \omega_{EM} \Delta T$ [36]. The phase is averaged over a time corresponding to half the gravitational wave's period, so $\Delta T = \frac{1}{2f_{\text{GW}}}$. The energy used in the measurement is $E = I_0 \frac{1}{2f_{\text{GW}}} = \hbar \omega_{EM} N$, where I_0 is the power of the laser and N is the number of photons detected in the averaging time. As the

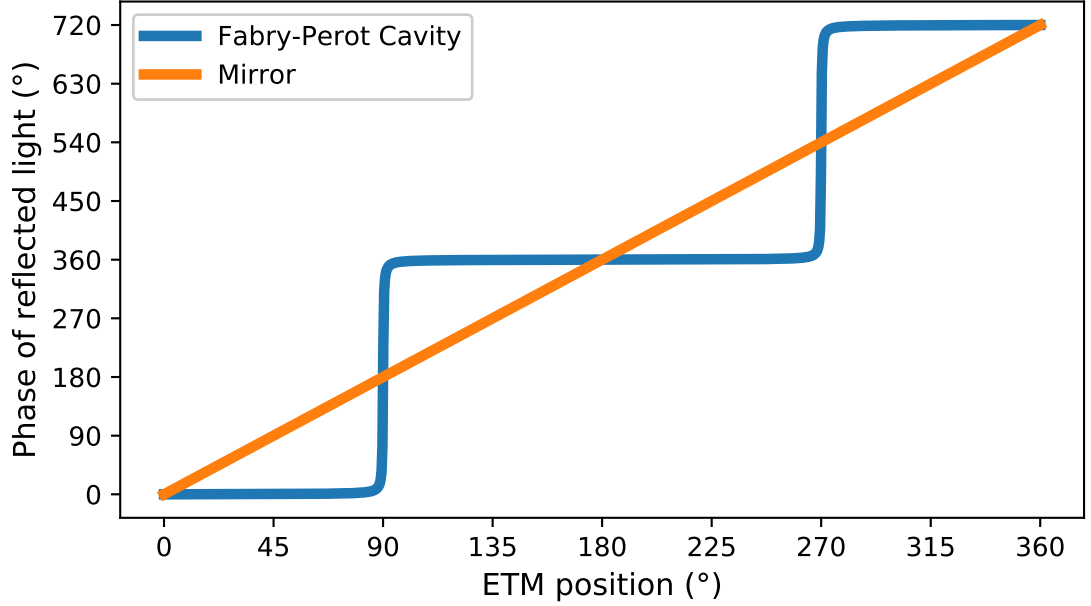


Figure 1.2: Comparison between the phase of the light reflected from a mirror versus the phase of light reflected from a cavity. The blue line shows the phase of the light that is reflected by a Fabry-Perot cavity. Initially, the cavity is on anti-resonance so perfect destructive interference occurs as the round-trip phase is 180° . The end mirror (ETM) is moved 90° , resulting in an extra 180° round-trip phase, and so the cavity resonates. The phase of the light reflected from the cavity rapidly changes as it moves through the resonance. The ETM is moved further, the cavity stops being resonant and the phase of the light is not changed as its reflected by the cavity. The orange line shows the phase of the light when a mirror is used instead of a cavity. The interferometers sensitivity is proportional to the slop $\frac{d\phi_{EM}}{d\phi_{ETM}}$, and so by inserting another mirror in the arm near the beam splitter, a Fabry-Perot cavity is formed and the sensitivity of the interferometer is increased.

laser is in a coherent state (e.g. [37–39]), the uncertainty detecting N photons is $\Delta N = \sqrt{N}$; therefore, the uncertainty in the energy used in the measurement is $\Delta E = \hbar\omega_{EM}\sqrt{N}$. Using the expressions for ΔE and $\Delta\phi$, the uncertainty principle may be used to obtain

$$N \geq \frac{1}{4(\Delta\phi)^2}. \quad (1.10)$$

The amplitude spectrum of the intensity measured by the photodiode over a time corresponding to a bandwidth of Δf at the antisymmetric port will depend on the amount of light at the dark port,

$$I_{\text{ASrms}} = \sqrt{2\overline{I_{\text{AS}}}\hbar\omega_{EM}\Delta f}. \quad (1.11)$$

The conversion from light power at the antisymmetric port to gravitational wave strain can be inferred from Equation (1.8),

$$\frac{h}{I_{\text{AS}}} = \frac{1}{2\sqrt{I_0\overline{I_{\text{AS}}}}}\frac{\lambda_{\text{EM}}}{4\pi LB}, \quad (1.12)$$

and so the amplitude spectrum of the strain corresponding to the shot noise of the light at the AS port is

$$I_{\text{ASrms}} = \sqrt{\frac{\hbar c^2}{8I_0\omega_{EM}L^2B^2}\Delta f}. \quad (1.13)$$

Equation 1.13 assumed the frequency response of the interferometer to a gravitational wave is flat, however the arm cavities have a low pass filtering effect on the signal, and so the shot noise will rise as f above the characteristic frequency of the arm cavity. This can be seen in the spectral density of the LIGO detector’s noise (Figure 1.3). Equation (1.13) is more rigorously derived in [34].

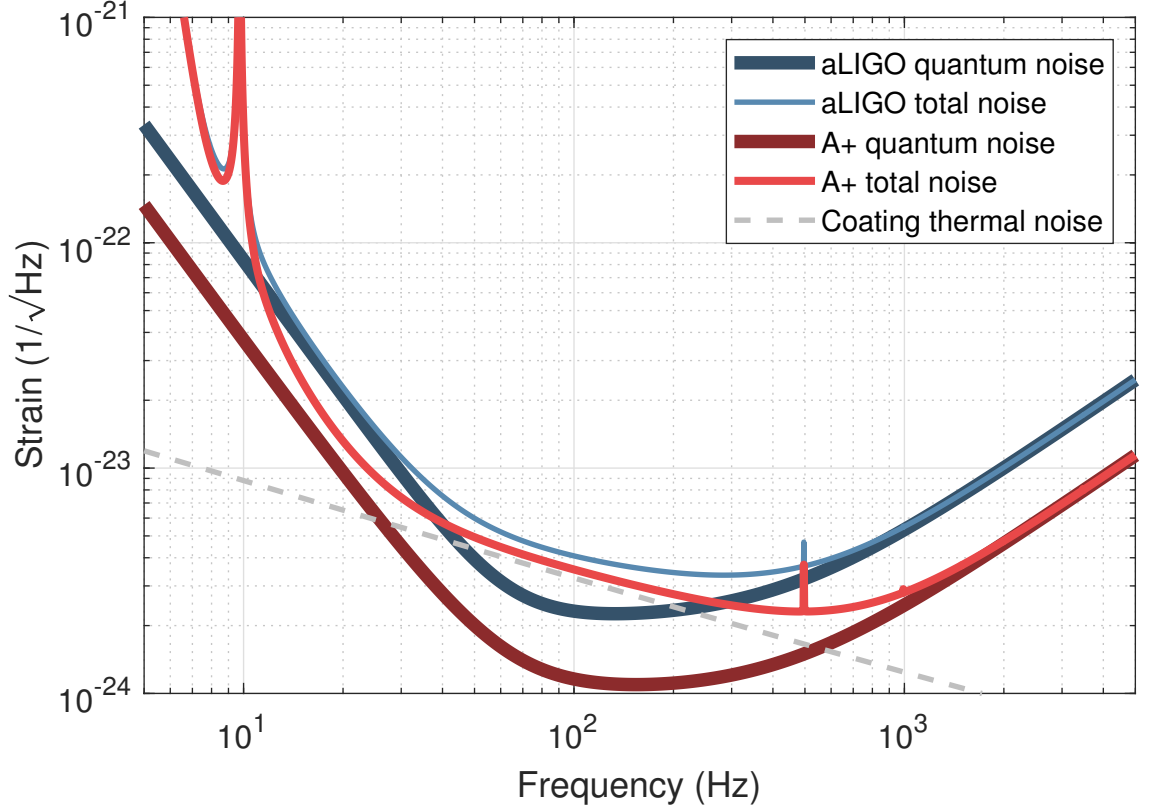


Figure 1.3: The strain Sensitivity of the aLIGO and A+ detectors. The noise due to radiation pressure and photon shot noise is shown in dark-blue (aLIGO) and dark-red (A+). Above 10 Hz, quantum effects are one of the limiting sources of noise; with the other being the noise due to Brownian motion of the mirror's coating. Below 10 Hz, seismic noise begins to dominate the noise of the detector. Above 100 Hz, shot noise dominates the noise of the detector. This rises with f as the shot noise is filtered by the arm cavities. Below 100 Hz, the quantum noise goes as $1/f$ due to radiation pressure.

To beat the photon shot noise by a factor of 10, Equation (1.9) and Equation (1.10) show that, for a LIGO-like interferometer, the required number of photons to measure a gravitational wave with a strain amplitude of $h_{\text{rms}} = 10^{-23}$ is $\sim 10^{22}$. For a gravitational wave signal with a frequency of 100 Hz, the laser power incident on the beam splitter would need to be $\sim 1\text{MW}$. No such laser exists.

1.4.2 Using Power Recycling to Reduce Shot Noise

To increase the laser power that is incident upon the beam splitter, a power recycling cavity can be used [40–42]. When the interferometer is on the dark fringe, most of the light from the laser gets reflected by it. Thus, the interferometer is, effectively, a mirror. Drever [43] [can't find a copy!](#) proposed that a *power recycling mirror* (PRM) could be placed between the laser and the interferometer to form a cavity. A sketch of this is shown in Figure 1.1. The total power impinging on the beam splitter would be increased by making wavelength scale position changes of, i.e. *tuning*, the position of the PRM such that the light reflected from the interferometer and the light from the laser add in phase, i.e. the reflected light power would be ‘recycled’. Current laser sources that can be operated for year-long timescales are capable of producing of the order 100W, see Chapter ??, and so the power recycling factor needs to be $\sim 10^4$. This technique has been implemented at LIGO [44] and other large-scale interferometers [26, 27].

1.4.3 Using Signal Recycling to Shape Quantum noise

The interferometer’s bandwidth depends on the amount of time that signal-carrying light is in the arms for. Meers suggested that the signal light could be recycled in

a similar fashion to how power is recycled [40]. This detection bandwidth can be altered by introducing a *signal recycling mirror* (SRM). A coupled cavity, known as the *signal recycling cavity* (SRC), between the SRM, ITM and ETM will be formed by placing the SRM between the photodiode and the beam splitter. A sketch of this is shown in Figure 1.1. The finesse of the arm cavities as seen by the light emerging at the antisymmetric port depends on the tuning of the SRM. There are two common modes of operation: *resonant sideband extraction* (RSE) [45, 46] and *signal recycling*. Alternatively, the SRM can be tuned so that the interferometer is optimised for a narrow band of frequencies to increase its sensitivity to certain BNS signals, e.g. [47].

When in RSE mode the SRM is tuned so that the SRC is anti-resonant, thus broadening the interferometers sensitivity bucket at the expense of losing peak sensitivity. This was motivated by needing to reduce the thermal load on the beam splitter by increasing the finesse of the arm cavities [46]. However, by increasing the finesse of the arm cavity, the bandwidth of the interferometer is reduced. To retain the desired arm cavity finesse from a sensitivity point of view, the SRM can be tuned so that the finesse of the arm cavity experienced by the signal is lowered.

In signal recycling mode, the SRM is tuned so that the SRC is resonant, deepening the sensitivity bucket at the expense of bandwidth. The trade-off between bandwidth and peak sensitivity comes from the conservation of energy. From an energy argument, the sensitivity of a detector to gravitational waves is determined by the detectors bandwidth and the energy within this bandwidth; so for the same amount of power at the beam splitter, this can be thought of as a fundamental limit

to the interferometer's sensitivity. This is known as the Mizuno limit [32].

1.4.4 Reducing Shot Noise by Removing Junk Light with a Mode Cleaner Cavity.

All light that is incident on the output photodiode contributes to the total shot noise of the measurement; therefore, only light which carries signal should be incident on the photodiode to enhance the SNR of the measurement. In an ideal interferometer, the light emerging at the AS port will be a beam that is defined by the mode that is resonant within the arm cavity. To minimise diffraction losses and due to the mirrors's manufacturing tolerances, the light in the arm cavity will be in the *fundamental Gaussian mode*. However, due to optical imperfections, e.g. astigmatism, and the RF side-bands used to control the interferometer, the light emerging from the AS port contains light in higher order modes. This undesired light is referred to as *junk light*.

To remove junk light, mode cleaners are used (e.g. [48, 49]). A mode cleaner is a high finesse cavity that strips away higher order mode light and the control sidebands. As higher order modes acquire a different amount of phase as they propagate compared to the fundamental Gaussian mode, an optical cavity can be designed such that the fundamental mode resonates while the higher order modes do not. These higher order modes will be suppressed by a factor of the cavity's finesse. However, it is possible for some higher order modes to not be suppressed if their round trip Gouy phase is nearly the same as that of the fundamental mode; thus, the cavity is designed so that only higher order modes which carry little light will do this. The control sidebands are attenuated if they are outwith the cavities

linewidth. The beam transmitted by the mode cleaner will be predominantly in the fundamental mode, and so the SNR of the signal measured by the photodiode will be increased.

1.4.5 Radiation Pressure Noise and the Standard Quantum Limit

Radiation pressure noise is a quantum effect that arises due to the uncertainty in the momentum that light imparts to the test masses when it is reflected from them [37]. These quantum fluctuations will give rise to a differential force on the ETMs, meaning that this noise couples as $1/f^2$ to the measurement of the positions of the ETMs. As the force exerted by a beam is $F = \hbar\omega_{EM}N/c = P/c$, the amplitude spectral density due to the radiation pressure of the length of a Fabry-Perot cavity with mirrors of mass m is

$$\Delta x_{\text{RPNFP}} = \frac{\sqrt{\hbar\omega_{EM}P}}{2mc\pi^2 f^2}. \quad (1.14)$$

To reduce the effect of radiation pressure noise, the LIGO test masses are 40 kg. The effect of radiation pressure noise on the LIGO detectors's sensitivity curves can be seen in Figure 1.3.

Radiation pressure noise scales with the quantum fluctuations entering the anti-symmetric port whereas shot noise is inversely proportional to these fluctuations. For a laser which produces a coherent state, i.e. a classical laser, the standard quantum limit (SQL) [37, 50, 51] sets an optimal laser power for a given ETM mass which minimises the total quantum noise. The SQL results in a $1/f$ locus that is

determined by the mass of the ETM and the duration of the measurement.

The sensitivity curve of an interferometer with a signal recycling mirror that is tuned to either signal recycling mode or RSE mode will ‘touch’ the SQL at one frequency; however, for an arbitrary detuning of the SRM, the sensitivity curve of the interferometer will drop below the SQL at some frequencies due to non-linear optical effects [52].

1.4.6 Coating Thermal Noise and Other Noise Sources

There are many classical sources of noise to overcome in a gravitational wave interferometer. The most notable of these is the noise associated with the Brownian motion of atoms in the amorphous mirror coating layers of the test masses [53]. Current detectors are designed around the expected noise level that can be achieved with these coatings. This will be one of the limiting sources of noise in the A+ detector, as shown in Figure 1.3. Other noise sources include seismic noise from the motion of the Earth, Newtonian noise from pressure waves in the Earth’s crust, residual gas pressure noise that alters the refractive index of the arms, and the noise associated with the interferometer’s electronics (e.g. [54, 55]).

1.5 Reducing Shot Noise with Non-Classical States of Light

The two-photon formalism developed by Caves and Schumaker [56, 57] can be used to see how non-classical states of light can improve the sensitivity of an interferometer. In this framework, light is analogous to a quantum simple harmonic oscillator (SHO).

A SHO can be described by two orthogonal components, such as the momentum and position quadrature; for light a convenient basis to use is the *amplitude* and *phase* quadrature. The amplitude and phase quadratures, $\hat{\mathbf{X}}$ and $\hat{\mathbf{Y}}$ respectively, defined in terms of annihilation and creation operators of the electromagnetic field, $\hat{a}$ and $\hat{a}^\dagger$, are (N.B repetition with BHD chapter)

$$\hat{\mathbf{X}}(t) = \frac{1}{2} (\hat{a}(t) + \hat{a}^\dagger(t)) , \quad (1.15)$$

$$\hat{\mathbf{Y}}(t) = \frac{1}{2i} (\hat{a}(t) - \hat{a}^\dagger(t)) . \quad (1.16)$$

From this formalism, an uncertainty relation between the phase and amplitude quadratures can be derived:

$$\Delta\hat{\mathbf{X}}\Delta\hat{\mathbf{Y}} \geq \frac{1}{2} \quad (1.17)$$

If the uncertainty in one quadrature is reduced, the uncertainty in the other is increased. Such a light field is *squeezed*. The phase between the squeezed field and the detected quadrature is defined as the squeezing angle.

As the gravitational wave signals that are to be measured will be in the phase quadrature of the light, it would be advantageous to, by the uncertainty principle, move noise to the undetected amplitude quadrature from the detected phase quadrature; this can be achieved by using squeezed light. Squeezed light is created with an optical parametric amplifier; this is a device which generates correlated photons by converting single higher energy photons into two lower energy photons. This squeezed vacuum light is injected into the interferometer from the AS port so that the observed noise within the interferometer is squeezed. This technique has been implemented at the LIGO Livingston Observatory (LLO) [58] and GEO600 [59].

However, fluctuations in the amplitude quadrature of the light perturb the position of the test masses; thus, they couple to the signal at the antisymmetric port. During LIGO’s third observation run, the squeezing angle had to be tuned to prevent radiation pressure noise from limiting the BNS range of the detector [58].

To infer the quantum state of a light field, a technique known as *balanced homodyne detection* (BHD) can be used to observe one quadrature of the light field [60]. By combining the signal light field that is to be measured with a much stronger *local oscillator* light field on a beam splitter, the quantum state of the signal light field is encoded in the combined beam’s intensity. The relative phase between the signal and local oscillator determines the quadrature of the quantum state that will be observed. Chapter ?? explores this in more detail.

1.6 Interferometry Techniques to Beat Quantum Limits

Squeezed light and an effect known as ponderomotive squeezing [50], can be used to surpass the SQL at some frequencies. Ponderomotive squeezing arises due to the natural correlations between the phase and amplitude quadratures of the light created by the interferometer. A demonstration of the SQL being surpassed at the LIGO Livingston detector is found in [61]. Although [61] shows the SQL was beaten at some frequencies, the noise at other frequencies was increased relative to if no squeezing was used. As the LIGO detectors are optimised for sensing gravitational waves, beating the SQL is not the main objective when devising interferometer configurations. However, [61] shows that the 40 kg test masses do behave quantum-

mechanically. Additionally, sub-SQL interferometer configurations are optimised for detecting certain astrophysical sources with merging frequencies around the frequency at which the SQL is beaten.

To avoid the increase in radiation pressure noise that is experienced when squeezed light is used, a frequency-dependent squeezer can be used [50]. The light's squeezing angle can be rotated as a function of frequency. This is done by using a high finesse detuned filter cavity. At frequencies below the linewidth of the cavity, the light reflected from the filter cavity will experience a phase shift and so the squeezing angle will be rotated; in contrast, at frequencies above the cavity's linewidth, the light incident upon the cavity is reflected without experiencing any phase shift and so the squeezing angle is not rotated. At low frequencies, the amplitude quadrature can be squeezed and the radiation pressure noise can be decreased. At high frequency, the phase quadrature can be squeezed and the shot noise can be decreased. A Frequency dependent squeezed light source that is suitable for LIGO has demonstrated [62] and frequency dependent squeezing will be part of the A+ detector.

Variational readout is a technique where quantum noise due to radiation pressure is removed from the measurement by using a frequency dependent homodyne angle [50]. This is realised in a similar way to frequency dependent squeezing; the AS beam is coupled to a filter cavity and thus the detected quadrature of the beam emerging from this cavity is frequency dependent. This beam is then measured on photodiodes using a conventional BHD scheme. Variational readout schemes are sensitive to small amounts of loss within the filter cavity, and so this technology is not mature enough to be implemented into current gravitational wave detectors.

To beat the bandwidth limit, energy has to be injected into the system. White light cavities as a means of increasing the detector bandwidth have been studied [63–65]. However, these interferometer configurations have yet to be experimentally demonstrated.

There are several sub-SQL schemes which avoid the quantum noise problems related to making a position measurement of the test masses. If a momentum measurement of the test masses could be made, then a random impartation of ‘position’ would not affect future measurements of the momentum [66]. Such measurements are known as quantum non-demolition. As a speed measurement is being made, the displacement noise would rise as $1/f$ instead of $1/f^2$; this type of measurement can be made with a speed meter interferometer [67–69].

1.7 The Effect Losses Have on Measuring Quantum States of Light

To preserve the benefits gained by using quantum states of light, it is essential to have minimal losses in the read-out of a gravitational wave detector. Real optical components have physical attributes that result in incident light power being wasted. These mechanisms are known as losses, and examples of these include absorption and scatter. A lossy optical component can be modelled as an ideal one downstream of a beam splitter whose splitting ratio represents the loss (e.g. [38, Section 2]). The incoming light enters side of the beam splitter, some light is directed to the ideal component and some gets lost. In the remaining input to the beam splitter, vacuum fluctuations can enter. Some of these fluctuations will make it to the ideal

component, and so the state of the light that is detected is contaminated with extra vacuum fluctuations compared to if there was no loss.

Therefore, it is critical that the photodiode used to measure the AS beam has low loss. Photodiodes convert photons into electrons, ideally with a one-to-one ratio. The measure of how well this is done is the photodiode's *quantum efficiency*, η . The factor by which the shot noise increases, ϵ , due to a simple loss mechanism, such as a non-ideal photodiode, is

$$\epsilon = \sqrt{\frac{1}{\eta} - 1}. \quad (1.18)$$

The smallest photodiode quantum efficiency acceptable in a gravitational wave interferometer is 99% [70]. This can be achieved by photodiodes fabricated from InGaAs.

Mode matching losses have a greater contribution to the total shot noise compared to losses of the same size generated by a simpler mechanism [71]. This source of loss can be reduced through careful design of the optical layout. Mirrors with a tunable radius of curvature can be used to ensure that there is minimal mode matching loss. This is the focus of Chapter ??.

1.8 Overview of the LIGO Detector

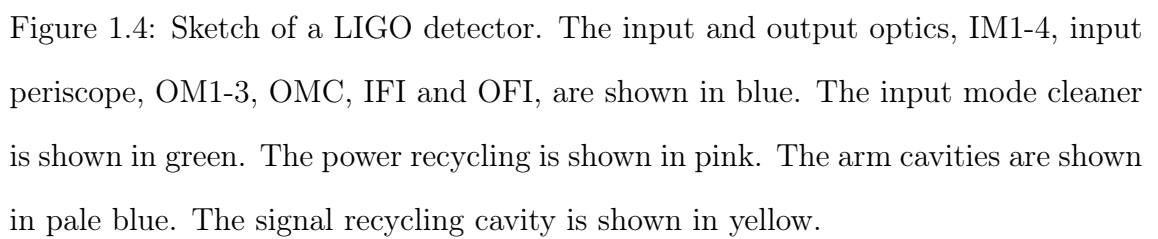
The LIGO detector is a dual-recycled-Fabry-Perot Michelson interferometer with 4 km arms [2]. The layout of the detector is shown in Figure 1.4 [72] and its sensitivity curve is shown in Figure 1.3 [73, 74]. The term dual-recycled refers to the fact that the interferometer uses both power and, the mathematical dual of power

recycling, signal recycling. The interferometer is housed within a vacuum system comprised of several horizontal access modules (HAM) and beam splitter chambers (BSC). Figure 1.4 includes all the optics from the pre-stabilised-laser (PSL) to the readout photodiodes that the main beam interacts with.

The first cavity after the PSL is the input mode cleaner; this provides a beam with high spatial purity to the interferometer and is used in the initial locking sequence as a stable reference frequency. The power recycling cavity features two extra mirrors, known as PR2 and PR3, in addition to the PRM. The role of these mirrors is to make round trip gouy phase sufficient so that it is stable; the signal recycling cavity also has three mirrors for the same reason [44]. The beam passes through the output Faraday isolator (OFI), and is steered into the output mode cleaner. The gravitational signal is encoded in the beam transmitted by the OMC, so photodiodes monitor the light transmitted by the OMC.

1.9 The Future of Ground Based Gravitational Wave Detectors

The next generation of ground-based gravitational wave detectors include the Einstein Telescope [75], LIGO Voyager [76] and LIGO Cosmic Explorer [77]. LIGO Voyager will be a cryogenic interferometer installed in the current LIGO infrastructure and will be a factor of ~ 2 more sensitive than LIGO A+. The Einstein telescope Cosmic explorer interferometers will be ten times larger than the current generation of interferometers and will be able to test the no-hair theorem, test alternative theories of gravity, test models of string theory, measure objects with large



redshifts to learn about the formation of cosmological structures, measure closer to the big bang than is possible with electromagnetic telescopes, detect cosmic strings, probe the physics of neutron stars and measure what happens during a supernova. As well as this, the third generation of gravitational wave detectors may measure signals from exotic sources which have not yet been theorised.

As the limiting sources of noise in current gravitational wave detectors is the coating thermal noise, research into cryogenic mirror coatings is under way [78]. Fused silica is opaque at these frequencies and its noise characteristics do not improve when cooled, so silicon test masses are being developed [76].

It is expected that low noise, low loss coatings will not be compatible with 1 μm light, so these detectors will operate with either a 1.5 μm or 2 μm wavelength laser [76]. Laser technology for 2 μm light is still in its infancy, and so there is lots of uncertainty in the practicality of using 2 μm as the operating wavelength of a gravitational wave detector; for instance, low noise, high quantum efficiency photodiodes that can detect 2 μm light have never been fabricated. This is discussed in Chapter ??.

1.10 Conclusion

Further detections of gravitational waves will lead to a better understanding of the universe. To reach the design sensitivity of the LIGO detectors, among other upgrades, the input laser power needs increasing. To further lower the LIGO detectors noise floor, the A+ upgrade will utilise frequency dependent squeezed light. For this to be successfully implemented, the quantum state of the light field must be

accurately detected with a balanced homodyne readout and losses in the read-out must be minimal.

As there is potential to reduce the noise floor that will limit the A+ detector by using cryogenic mirror coatings that are compatible with $2\text{ }\mu\text{m}$ light, future detectors may use a $2\text{ }\mu\text{m}$ laser; this leads to the question: is laser technology at this wavelength sufficiently mature enough to enable quantum limited measurements? Balanced homodyne detection and photodiodes with high quantum efficiency and low dark noise will be required for any future interferometer.

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